Homework 6

Michelle Arendt, Jiawen Long, Sijie Liu

Commit 1

Our scripts file is named "hw6scripts"; the data file is "hw6data."

Commit 2

This homework was completed using R (R Core Team 2026). The tidyverse package (Wickham et al. 2019) provides tools for data manipulation and visualization, while the knitr package (Xie 2015) was used for reproducible reporting.

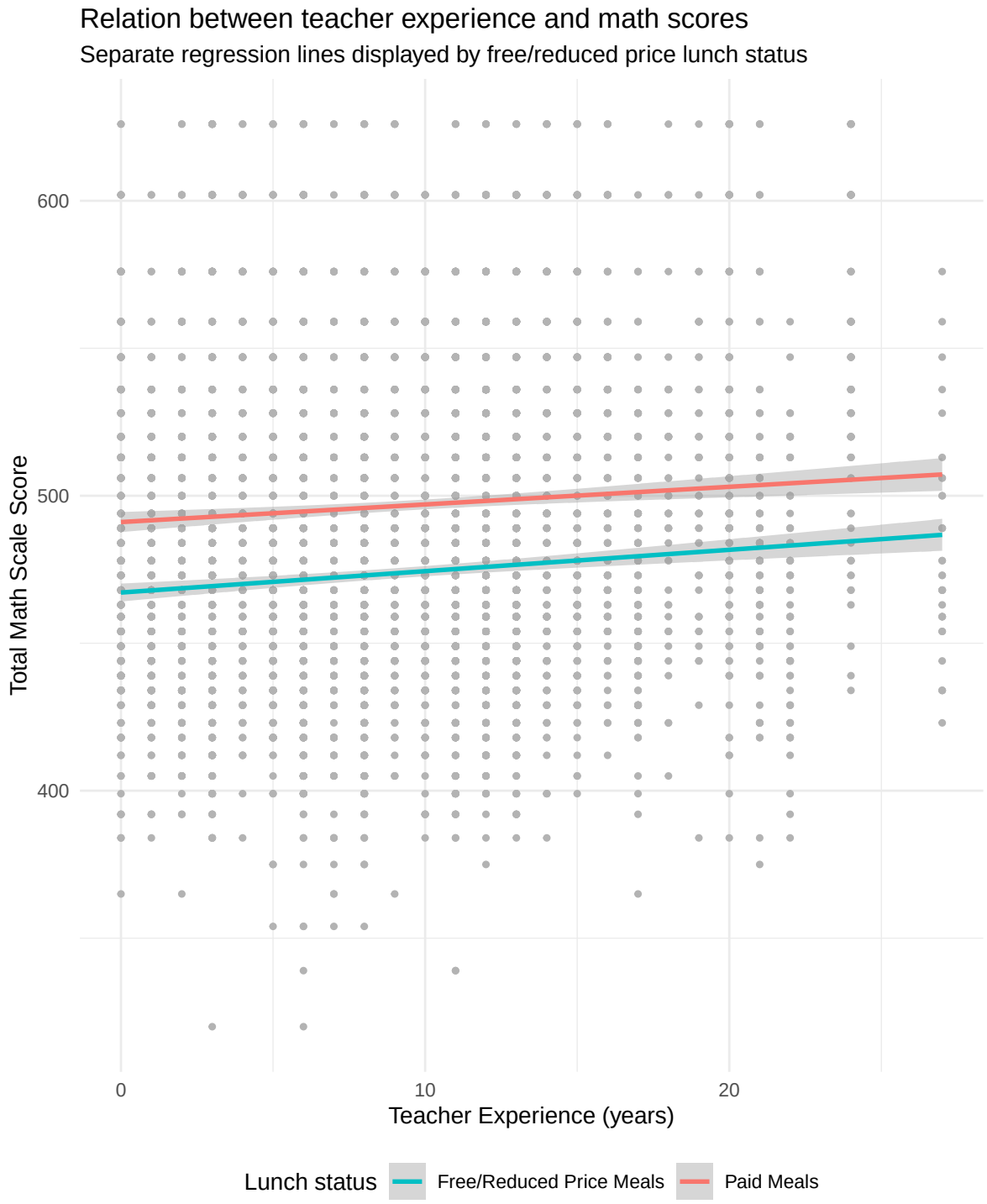
We read an article by Nese (2022), which examined the growth and predictive performance of oral reading fluency measures. Two other articles were published in the same journal. One examined racial-ethnic inequality in school access (Speroni et al. 2026) and another investigated ethnic-racial identity projects in Brazil (Callegari et al. 2026).

Commit 3

sex	frl	math_mean	math_sd	rdg_mean	rdg_sd
boy	no	492.85	46.34	441.46	32.32
boy	yes	469.87	46.09	425.38	26.63
girl	no	501.21	45.96	448.54	34.52
girl	yes	477.51	46.30	430.80	27.42

The table shows the mean and standard deviation of math and reading scores by sex and FRL status

Commit 4


Relation between teacher experience and math scores
Separate regression lines displayed by free/reduced price lunch status
Total Math Scale Score
600
500
400
0
10
20
Teacher Experience (years)
Lunch status
Free/Reduced Price Meals
Paid Meals

The plot shows the relationship between teacher experience and math scores, with separate linear regression lines for students from families of different income levels (as indicated by their lunch status: receiving free/reduced price meals or not). The slopes for both regression lines are similarly positive, indicating that teacher experience has a comparable positive effect on math scores regardless of their household income condition. However, students coming from lower-income families (receiving free/reduced price meals) tend to have lower math scores overall compared to their peers who do not receive free/reduced price meals.

References

Callegari, Julia, Tabitha Bonilla, Guilherme Lichand, and Emma K. Adam. 2026. "The Presence of Ethnic-Racial Identity School Projects and Youth Development in Brazil." *AERA Open* 12: 23328584261499280. https://doi.org/10.1177/23328584261499280.

Nese, Joseph F. T. 2022. "Comparing the Growth and Predictive Performance of a Traditional Oral Reading Fluency Measure with an Experimental Novel Measure." *AERA Open* 8: 23328584211071112. https://doi.org/10.1177/23328584211071112.

R Core Team. 2026. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.R-project.org/.

Speroni, Samuel, Evelyn Blumenberg, and Sophia Balkovski. 2026. "Racial-Ethnic Inequality in Geographic Access to Choice Schools in California." *AERA Open* 12: 23328584261489190. https://doi.org/10.1177/23328584261489190.

Wickham, Hadley, Mara Averick, Jennifer Bryan, et al. 2019. "Welcome to the Tidyverse." *Journal of Open Source Software* 4 (43): 1686. https://doi.org/10.21105/joss.01686.

Xie, Yihui. 2015. *Dynamic Documents with r and Knitr*. 2nd ed. Chapman; Hall/CRC. https://yihui.org/knitr/.